Physical Chemistry of Materials

Lecture 05

Applications of Canonical Ensemble

Last Class

(Thermodynamic) Ensemble

Idealized group of a large number of virtual copies

(sometimes infinitely many) of a system

Microcanonical ensemble

Canonical ensemble

Grand canonical ensemble

⋮

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Microcanonical Ensemble

Thermodynamic ensemble of an isolated system

- Conserved quantities: N, V, U (or traditionally N, V, E)
- All microstates are equally probable
- Probability to find the system in microstate j

$$p_j = \frac{1}{W}$$

Last Class

Canonical Ensemble

Thermodynamic ensemble of a closed system

- Conserved quantities: N, V, T
- Microstates may have different energies
- All microstates are NOT equally probable
- Probability to find the system in some microstate?

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Principle of Maximum Entropy

The probability distribution which best represents the current state of knowledge about a system is the one with largest entropy

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Ensemble Energy Conservation

$$S/k_{\mathrm{B}} = -\sum_i p_i \ln p_i$$

$$\sum_i p_i - 1 = 0$$

We can introduce one more constraint: $\sum_i n_i E_i = \mathcal{N} U$

As the constraint implies $\sum_i p_i E_i - U = 0$,

$$\frac{\partial[S/k_{\mathrm{B}} - \lambda(\sum_i p_i - 1) - \beta(\sum_i p_i E_i - U)]}{\partial p_i} = 0$$

$$p_i = e^{-1-\lambda-\beta E_i}$$

Summary

For a canonical ensemble, the probability to find the system in a microstate i (with energy E_i) is

$$p_i = e^{-1-\lambda-\beta E_i} = \frac{e^{-\beta E_i}}{Z} = \frac{e^{-E_i/k_\mathrm{B}T}}{Z}$$

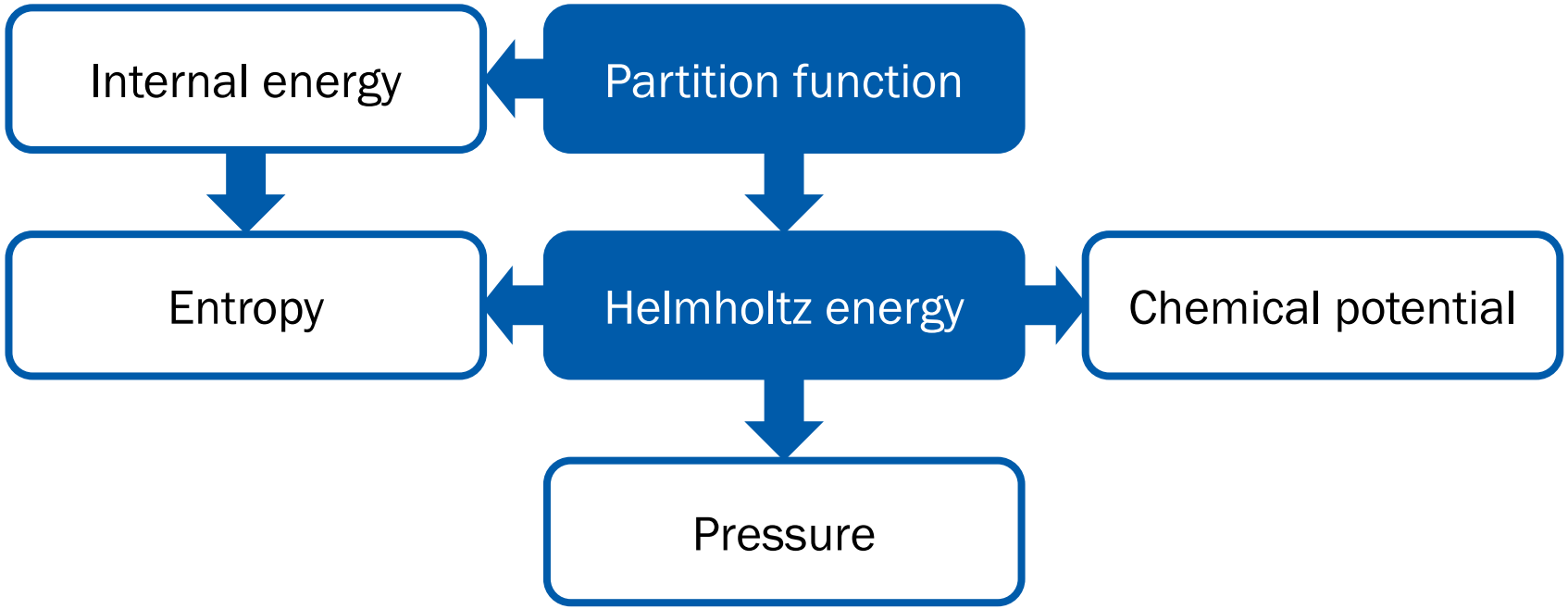
where the partition function $Z = \sum_i e^{-\beta E_i} = \sum_i e^{-E_i/k_\mathrm{B}T}$

Also, the Helmholtz energy is given by

$$A = -k_\mathrm{B}T \ln Z$$

Last Class

Z and Thermodynamics



Multi-Particle Partition Functions

- Independent & distinguishable particles

$$Z = z^N \text{ (}z\text{: one-particle partition function)}$$

- Independent & indistinguishable particles

$$Z = \frac{1}{N!} z^N$$

- Interacting particles: later

Two Distinguishable Particles

- Consider two particles A and B, with energies $\varepsilon_i^{\mathrm{A}}$ and $\varepsilon_j^{\mathrm{B}}$
- Partition function of A: $z_{\mathrm{A}} = \sum_i e^{-\beta \varepsilon_i^{\mathrm{A}}}$
- Partition function of B: $z_{\mathrm{B}} = \sum_j e^{-\beta \varepsilon_j^{\mathrm{B}}}$
- Total energy: $E_{ij} = \varepsilon_i^{\mathrm{A}} + \varepsilon_j^{\mathrm{B}}$
- Total partition function

$$Z = \sum_{i,j} e^{-\beta E_{ij}} = \sum_i \sum_j e^{-\beta\left(\varepsilon_i^{\mathrm{A}} + \varepsilon_j^{\mathrm{B}}\right)} = z_{\mathrm{A}} z_{\mathrm{B}}$$

N Distinguishable Particles

Similarly,

$$Z = z_{\mathrm{A}} z_{\mathrm{B}} \cdots z_{\mathrm{X}}$$

If the one-particle partition functions are identical ($= z$),

$$Z = z^N$$

Two Indistinguishable Particles

For two indistinguishable particles,

particle 1 in state A & particle 2 in state B

= particle 1 in state B & particle 2 in state A

Hence, the “correct” partition function is

$$Z = \frac{z_A z_B}{2}$$

N Indistinguishable Particles

We need to consider Gibbs' correction factor: $1/N!$

$$Z = \frac{1}{N!} z_{\mathrm{A}} z_{\mathrm{B}} \cdots z_{\mathrm{X}}$$

If the one-particle partition functions are identical ($= z$),

$$Z = \frac{1}{N!} z^N$$

Two-State Example

Consider a particle with an internal state: state A or state B.

$$E(\mathrm{A}) = \epsilon$$

$$E(\mathrm{B}) = -\epsilon$$

The one-particle partition function is

$$z = \sum_s e^{-\beta E(s)} = e^{-\beta E(\mathrm{A})} + e^{-\beta E(\mathrm{B})}$$

$$z = e^{-\beta\epsilon} + e^{+\beta\epsilon} = 2\cosh(\beta\epsilon)$$

Partition Function

$$z = e^{-\beta\epsilon} + e^{+\beta\epsilon} = 2\cosh(\beta\epsilon)$$

- Distinguishable particles:

$$Z = z^N = [2\cosh(\beta\epsilon)]^N$$

- Indistinguishable particles:

$$Z = \frac{1}{N!} z^N = \frac{1}{N!} [2\cosh(\beta\epsilon)]^N$$

Thermodynamics

- Helmholtz energy

$$A = -k_{\mathrm{B}} T \ln Z = -N k_{\mathrm{B}} T \ln[2\cosh(\beta\epsilon)]$$

- Internal energy

$$U = -\frac{\partial \ln Z}{\partial \beta} = -N \frac{\partial}{\partial \beta} \ln[2\cosh(\beta\epsilon)]$$

$$\therefore \; U = -N\epsilon \tanh(\beta\epsilon)$$

- Entropy: $S = (U - A)/T$

$$S = \frac{1}{T}\{-N\epsilon \tanh(\beta\epsilon) + N k_{\mathrm{B}} T \ln[2\cosh(\beta\epsilon)]\}$$

Classical Statistical Mechanics

Phase Space

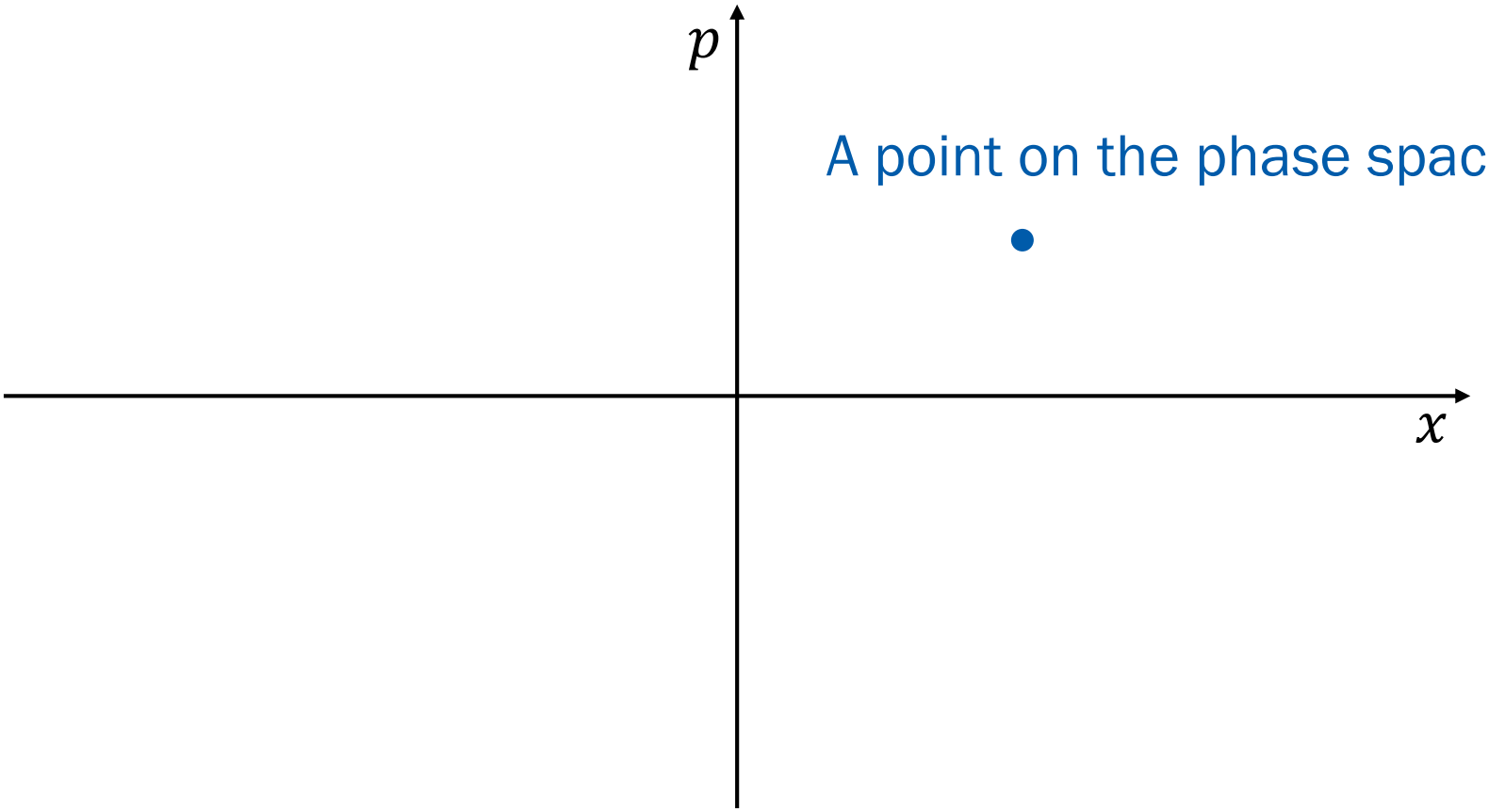
Space of all possible microstates of a system

- Classical mechanics:

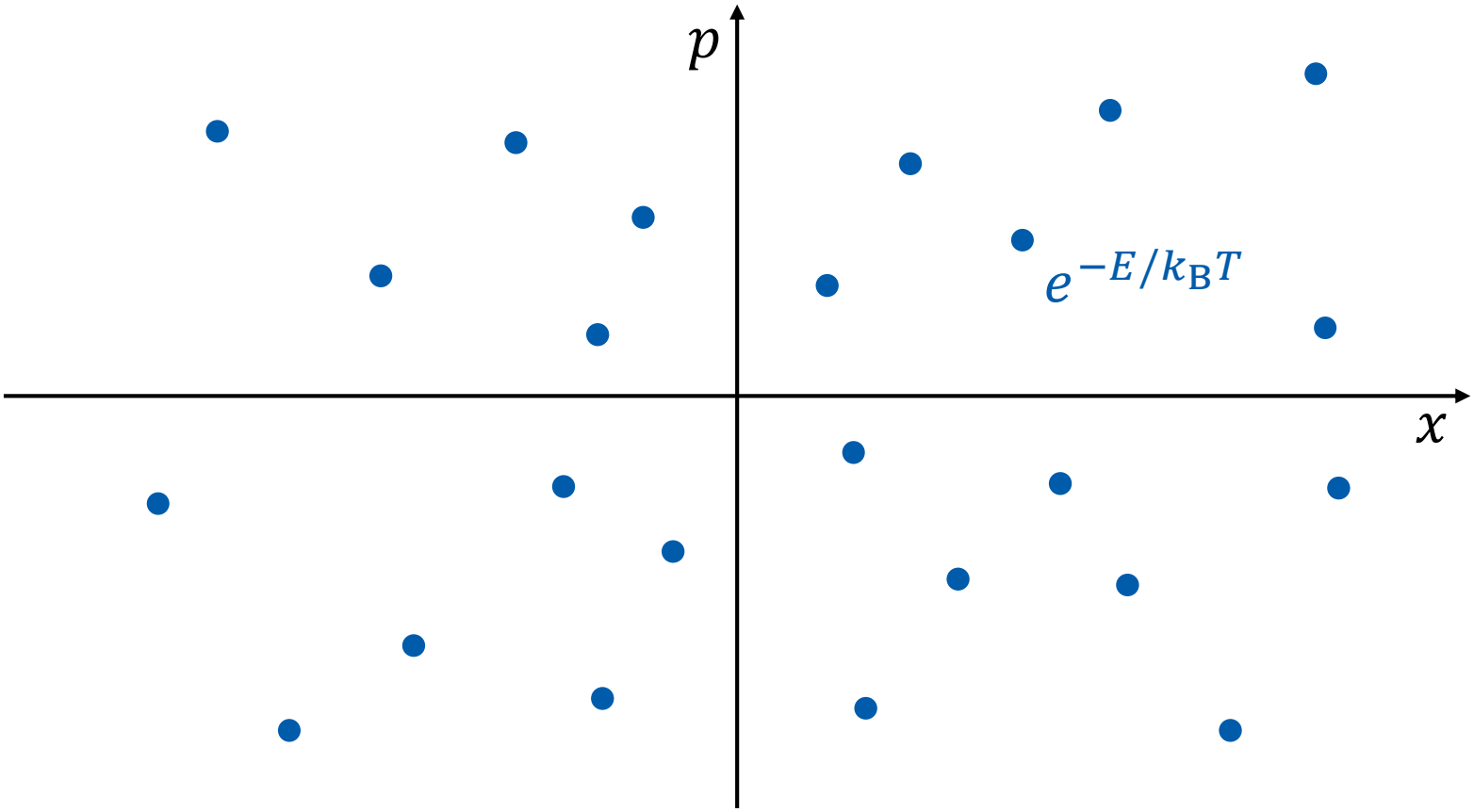
  Space of all possible values of

  position and momentum variables

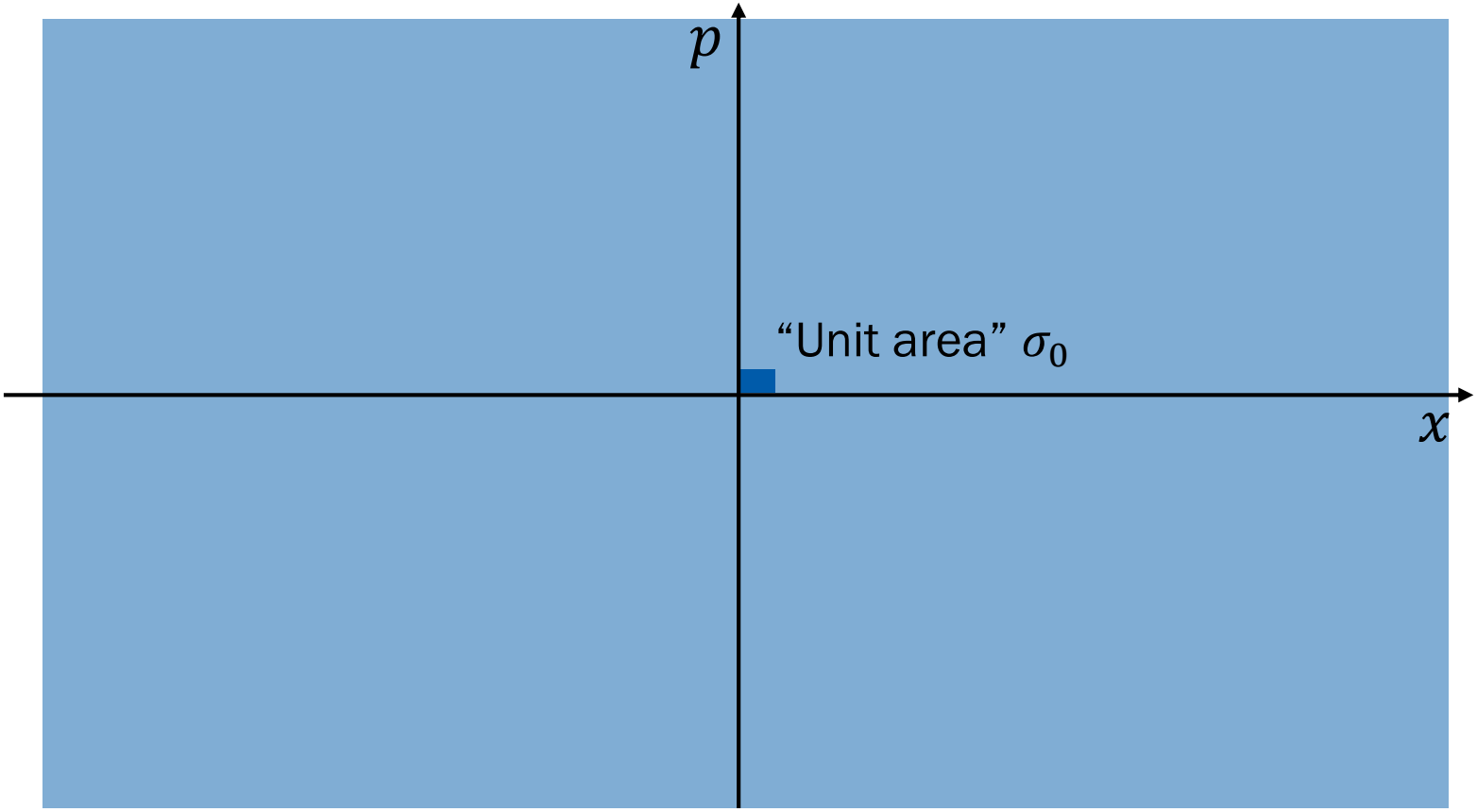
Microstate of Classical Particle



Classical Partition Function



Classical Partition Function



Classical Partition Function

$$z = \sum_s e^{-\beta E(s)} \quad \rightarrow \quad z = \frac{1}{\sigma_0} \int \int e^{-\beta E(x,p)} \, \mathrm{d}x \mathrm{d}p$$

for a 1-dimensional single particle

Two Classical Particles

- Consider two particles A and B, with energies ε^{A} and ε^{B}
- Partition function of A: $z_{\mathrm{A}} = \frac{1}{\sigma_0} \int e^{-\beta \varepsilon^{\mathrm{A}}} \, \mathrm{d}^3 q_{\mathrm{A}} \mathrm{d}^3 p_{\mathrm{A}}$
- Partition function of B: $z_{\mathrm{B}} = \frac{1}{\sigma_0} \int e^{-\beta \varepsilon^{\mathrm{B}}} \, \mathrm{d}^3 q_{\mathrm{B}} \mathrm{d}^3 p_{\mathrm{B}}$
- Total energy: $E = \varepsilon^{\mathrm{A}} + \varepsilon^{\mathrm{B}}$
- Total partition function

$$Z = \frac{1}{\sigma_0^2} \int e^{-\beta E} \, \mathrm{d}^3 q_{\mathrm{A}} \mathrm{d}^3 q_{\mathrm{B}} \mathrm{d}^3 p_{\mathrm{A}} \mathrm{d}^3 p_{\mathrm{B}} = z_{\mathrm{A}} z_{\mathrm{B}}$$

N Particles

- Distinguishable particles

$$Z = z_{\mathrm{A}} z_{\mathrm{B}} \cdots z_{\mathrm{X}}$$

If the one-particle partition functions are identical ($= z$),

$$Z = z^N$$

- Indistinguishable particles

$$Z = \frac{1}{N!} z^N$$

Ideal Gas (Classical Version)

The system consists of indistinguishable particles:

$$Z = \frac{1}{N!} z^N$$

Ideal Gas (Classical Version)

- Classical treatment of z:

$$z = \frac{1}{\sigma_0} \int \int e^{-\beta E(x,p)} \, \mathrm{d}x \mathrm{d}p$$

Since $E(x,p) = p^2/2m$,

$$z = \frac{1}{\sigma_0} \int \int e^{-\beta p^2/2m} \, \mathrm{d}x \mathrm{d}p$$

Partitioning of z

$$z = \frac{1}{\sigma_0} \int \int e^{-\beta p^2/2m} \, \mathrm{d}x \mathrm{d}p$$

$$z = \frac{1}{\sigma_0} \left[\int 1 \, \mathrm{d}x \right] \left[\int e^{-\beta p^2/2m} \, \mathrm{d}p \right]$$

$$\int 1 \, \mathrm{d}x = \int_0^a 1 \, \mathrm{d}x = a$$

Momentum Integral

$$z = \frac{a}{\sigma_0}\left[\int e^{-\beta p^2/2m}\,\mathrm{d}p\right]$$

$$\int e^{-\beta p^2/2m}\,\mathrm{d}p = \int_{-\infty}^{\infty} e^{-\beta p^2/2m}\,\mathrm{d}p$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2}\,dx = \sqrt{\pi/\alpha}$,

$$z = \frac{a}{\sigma_0}\left(\frac{2\pi m}{\beta}\right)^{1/2}$$

For N Particles

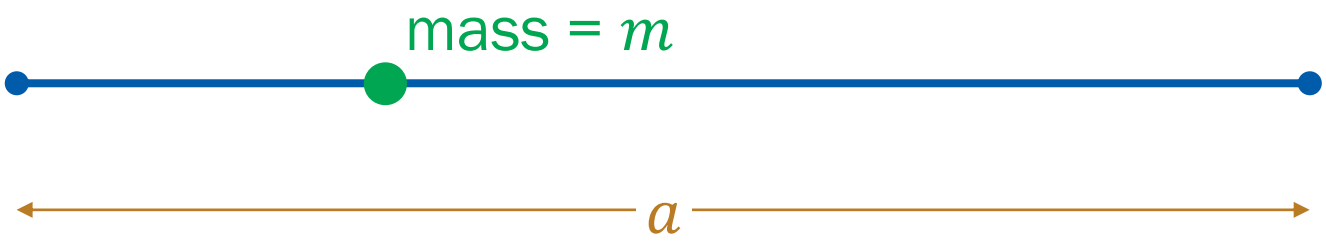
$$z = a\left(\frac{2\pi m}{\beta\sigma_0^2}\right)^{1/2}$$

$$Z = \frac{1}{N!}\left(\frac{2\pi m}{\beta\sigma_0^2}\right)^{N/2} a^N$$

For a 3-dimensional system,

$$Z = \frac{1}{N!}\left(\frac{2\pi m}{\beta\sigma_0^2}\right)^{3N/2} a^{3N} = \frac{1}{N!}\left(\frac{2\pi m}{\beta\sigma_0^2}\right)^{3N/2} V^N$$

Particle in a Box



$$\mathcal{V}(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 \le x \le a \\ \infty, & x > a \end{cases}$$

Review: Quantum Mechanics

- Quantum number $n = 1, 2, 3, \cdots$
- Energy levels

$$E = \frac{h^2}{8ma^2} n^2 = E_n$$

- Wavefunctions

$$\psi(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

Ideal Gas (Quantum Version)

The system consists of indistinguishable particles:

$$Z = \frac{1}{N!} z^N$$

- Quantum treatment of z:

$$E_n = \frac{h^2}{8ma^2} n^2 \ (n = 1, 2, 3, \cdots)$$

$$z = \sum_{n=1}^{\infty} e^{-\beta h^2 n^2 / 8ma^2}$$

Approximation

$$z = \sum_{n=1}^{\infty} e^{-\beta h^2 n^2 / 8ma^2}$$

Approximate

$$\sum_{n=1}^{\infty} e^{-\alpha n^2} \rightarrow \int_0^{\infty} e^{-\alpha n^2} \, \mathrm{d}n = \left(\frac{\pi}{4\alpha}\right)^{1/2}$$

Hence,

$$z = \left(\frac{\pi}{4\alpha}\right)^{1/2} = \left(\frac{2\pi m}{\beta h^2}\right)^{1/2} a$$

For N Particles

$$Z = \frac{1}{N!} z^N$$

$$Z = \frac{1}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{N/2} a^N$$

For a 3-dimensional system,

$$Z = \frac{1}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{3N/2} a^{3N} = \frac{1}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{3N/2} V^N$$

Comparison

- Classical version

$$Z = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{\sigma_0^2}\right)^{3N/2}$$

- Quantum version

$$Z = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}$$

$$\therefore \sigma_0 = h$$

Helmholtz Energy

$$Z = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}$$

Now, we can calculate the Helmholtz energy:

$$A = -k_\mathrm{B} T \ln Z = -k_\mathrm{B} T \ln\left\{\frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}\right\}$$

$$A = -N k_\mathrm{B} T \ln V + k_\mathrm{B} T \ln N! - \frac{3}{2} N k_\mathrm{B} T \ln\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)$$

Internal Energy

$$Z = \frac{V^N}{N!}\left(\frac{2\pi m k_B T}{h^2}\right)^{3N/2}$$

Also the internal energy:

$$U = -\frac{\partial(\ln Z)}{\partial\beta} = -\frac{\partial}{\partial\beta}\ln\left\{\frac{V^N}{N!}\left(\frac{2\pi m}{\beta h^2}\right)^{3N/2}\right\}$$

$$U = \frac{\partial}{\partial\beta}\ln\beta^{3N/2} + \frac{\partial}{\partial\beta}(\text{terms independent to } \beta) = \frac{3}{2}Nk_BT$$

Pressure

$$A = -Nk_{\mathrm{B}}T \ln V + k_{\mathrm{B}}T \ln N! - \frac{3}{2} Nk_{\mathrm{B}}T \ln\left(\frac{2\pi m k_{\mathrm{B}}T}{h^2}\right)$$

Using $p = -(\partial A/\partial V)_T$,

$$p = Nk_B T \frac{\partial(\ln V)}{\partial V} + \frac{\partial}{\partial V}(\text{terms independent to } V)$$

$$p = \frac{Nk_{\mathrm{B}}T}{V}$$

One can obtain entropy similarly